

ESTELLE JANIN

PhD Candidate – Extraterrestrial Life Detection

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🐙 GitHub

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INTERDISCIPLINARY RESEARCH

A New Framework for Detecting Life on Exoplanets

Ph.D. Dissertation - first project

📅 Aug. 2022 – Ongoing

📍 ASU

- Apply Assembly Theory to the chemistry of planetary atmospheres
- Metabolism-agnostic and sensitive to life-as-we-don't-know-it
- Understand how the complexity generated at a planetary-scale by a bio/technosphere influences measurable properties of the atmosphere and its observational spectrum
- Translate research insights into recommendations for future mission design and development, such as the Habitable Worlds Observatory
- Combine theoretical innovation and empirical grounding, at the intersection of exoplanet science, astrobiology and complex systems science
- Supervised by Sara Walker and Mike Line

Evolutionary Transitions in Language & Communication

Ph.D. Dissertation - second project

📅 Aug. 2022 – Ongoing

📍 ASU

- Apply Assembly Theory to the emergence and evolution of intelligent communication
- Investigate communication from emergent artificial agents, human language corpora, and animal communication
- Supervised by Sara Walker, Philip Mauskopf and Patrick Young

Planetary Atmospheres through Network Theory

Graduate Collaborative Project

📅 Aug. 2021 – Ongoing

📍 ASU

- Investigate the detectability of topological reaction networks in terrestrial atmospheres, and distinguish abiotic vs. biotic pathways
- With Tessa Fisher, supervised by Sara Walker

Life, Otherwise: Between Science & Art

Science Contractor for the Berggruen Institute

📅 June 2023 – Ongoing

📍 Remote

- Make the Strange Familiar: create a physically plausible planetary system – Proxima Kósmos – where each planet features a different imagined alien biosphere
- Large collaboration between scientists, anthropologists, writers, philosophers, artists and graphic designers, with outputs ranging from rigorous planetary simulations to essays, short stories and art pieces
- Supervised by Claire Webb

Towards the Habitable Worlds Observatory

Leader of a Biosignature Science Case

📅 Jan. 2024 – Ongoing

📍 Remote

- Co-lead author of a Science Case Document (SCDD) titled *Searching for Life As We Don't Know It*
- Active participant of the Red Limit working group, writing a white paper about theory-observations trade-offs in the infrared for a variety of science cases
- Participated in the Living Worlds working group (biosignature possibilities)

EDUCATION

Ph.D. Astrophysics

Supervised by Sara Walker

📅 2022 – Ongoing

📍 Arizona State University

- School of Earth and Space Exploration (SESE)
- **Current GPA: 4.0**

MSci Natural Sciences

Astrophysics and Molecular & Cell Biology

📅 2018 – 2022

📍 University College London

- First Class Honors (average: 85%)

French Scientific Baccalaureate

Highest Honours: 19.63/20

📅 2015 – 2018

📍 Lycée Hoche, France

TEACHING

SES 376/377

📅 Spring 2025

📍 Arizona State University

Communicating Astronomical and Planetary Sciences, with Prof. Karen Knierman

SESE Graduate Mentor

📅 2025-2026

📍 Arizona State University

Undergraduate Thesis Supervisor

📅 2023 – Ongoing

📍 Arizona State University

Cameron Brooks, Jonathan Rabinowitz

AWARDS & GRANTS

2026 Student Travel Grant

from the HWO Project Office to attend Ab-SciCon (\$1,000)

2026 Travel Grant Award

ASU GPSA (\$950)

2023 Origins of Life Scholarship

to attend the Complex Systems Summer School (\$5100)

2023 Career Development Award

ASU GPSA (\$950)

2022 ASU First Year Fellowship

funds for one year of research

2022 UCL Dean's List

top 5% of graduating students

2022 Tessella Prize

best use of software in final year project

2021 Best Natural Sciences Literature Report

Exoplanet Spectroscopy and Radiative Transfer

Intercomparison Leader and NASA Collaborator

- 📅 Aug. 2019 – Ongoing 📍 NASA GSFC & Remote
- Community-wide model intercomparison of radiative transfer tools for exoplanet atmospheres: leading the transit case study
- Part of the organizational committee of the BUFFET-2 workshop (exo-planet model intercomparison)
- Detectability of CO₂ in TRAPPIST-1e-like exoplanet atmospheres
- Supervised by Thomas Fauchez, Geronimo Villanueva and Ravi Kopparapu

Extraterrestrial Communication: A Firefly-inspired Study

Undergraduate Student Supervisor

- 📅 Jan. 2023 – Ongoing 📍 ASU
- Cameron Brooks's Honor's Thesis: A Firefly-Inspired Model to Extraterrestrial Communication; drawing inspiration from firefly communication to infer the structure of pulsar signals around Earth that would demonstrate evolutionary distinctiveness from the astrophysical background, informing alien intelligence detection
- Manuel Espinoza's undergraduate project: Mapping intra-species variability of firefly flash assembly to the corresponding genomic variability, investigating communicative flexibility and rigidity of encoding

Eclipse Mapping & Human Cultural Evolution

Undergraduate Student Supervisor

- 📅 Jan. 2025 – Apr. 2026 📍 ASU
- Jonathan Rabinowitz's Thesis: Solar Eclipses and their Connection to Human Cultures and Societies
- Reconstructing solar eclipse paths at scale and correlating celestial events with social dynamics across past civilizations

Exploratory Research Projects in Complexity Science

Complex Systems Summer School

- 📅 Summer 2023 📍 Santa Fe Institute
- Simulating the life-cycle of scientific theories
- The structure of Abstraction: Word embeddings and Poincaré maps
- Oscillatory forcing in predicting civil strife events

Chemical Disequilibrium in Exoplanet Atmospheres

MAPS Summer Internship

- 📅 June-July 2022 📍 UCL
- Test and improve a chemical disequilibrium code (FRECKLL) for atmospheric models and retrievals (TauREx)
- Supervised by Giovanna Tinetti and Quentin Changeat

Modelling Transitional Planets: From Planetary Formation to Atmospheric Characterisation

Master's Thesis

- 📅 Oct. 2021 – May 2022 📍 UCL
- TauREx-enabled population study of 18 exoplanets, investigating the atmospheric properties and detectability of Water Worlds
- Supervised by Giovanna Tinetti and Quentin Changeat, with support from Tadahiro Kimura (NAOJ, Japan)

Exoplanet Biosignatures: Past, Present and Future

Undergraduate Dissertation

- 📅 Sept. 2020 – June 2021 📍 UCL

RESEARCH VISITS

University of Glasgow

- 📅 July 2024 📍 Glasgow, UK

invited by Lee Cronin

Earth-Life Science Institute

- 📅 Sept. 2023 📍 Tokyo, Japan

invited by Harrison Smith

Space Telescope Science Institute

- 📅 May 2023 📍 Baltimore, MD, USA

invited by Quentin Changeat

NASA Goddard Space Flight Center

- 📅 Summer 2022 📍 Greenbelt, MD, US

invited by Thomas Fauchez

WORKSHOPS & CONF.

AbSciCon

- 📅 May 2026 📍 Madison, WI, USA
- 📅 May 2024 📍 Providence, RI, USA

EvoLang

- 📅 Apr. 2026 📍 Plovdiv, BG
- 📅 May 2024 📍 Madison, WI, USA

AASTCS 11: Exoplanet Atmospheres 2026

- 📅 Mar. 2026 📍 Denver, CO, USA

American Astronomical Society meetings

- 📅 Jan. 2026 📍 Phoenix, AZ, USA
- 📅 June 2025 📍 Anchorage, AK, USA

Towards the Habitable Worlds Observatory

- 📅 July 2025 📍 Washington DC, USA

Schmidt Interdisciplinary Science Summit

- 📅 July 2025 📍 Pasadena, CA, USA

Building a Unified Framework For Exoplanet Treatments (BUFFET)

- 📅 Oct. 2024 📍 Remote
- 📅 Oct. 2023 📍 Remote
- 📅 Oct. 2022 📍 Remote
- 📅 Sept. 2021 📍 Remote

Exoplanets 5

- 📅 June 2024 📍 Leiden, NL

Habitable Worlds Observatory START-TAG meeting

- 📅 June. 2024 📍 Baltimore, MD, USA

Communicating Discoveries in the Search for Life in the Universe

- 📅 March 2024 📍 remote

Society for Literature, Science and the Arts - ALIEN

- 📅 Oct. 2023 📍 Phoenix, AZ, USA

- Thorough review and critical assessment of the evolution of the field, its current status and its complex interdisciplinary feedbacks
- Supervised by Giovanna Tinetti

International University Mars Rover Challenge

Leader of the UCL Science Team

📅 2019 – 2022

📍 UCL

- Designing and building a rover with an on-board science cache able to discriminate between extant, extinct and absent life on the Martian surface

PUBLICATIONS

📄 Lead/Co-Lead Author

- Janin, Estelle. (2021), "Exoplanet biosignatures", *Astronomy & Geophysics*, 62(6), 6.26–6.33
- Walker, S., Janin, E. et al. (2025), "Searching for Life-As-We-Don't-Know-It: Mission-relevant Application of Assembly Theory for Exoplanet Life Detection". NOI as a response to NASA-DARES 2025 RFI. arXiv: 2603.11086
- Janin, Estelle. (2022), "Planetary Habitability.", *Observatory*, 142(1290), 239-240.
- **in prep:** Janin E. et al., "Detecting Exoplanet Biospheres and Technospheres using Assembly Theory and Infrared Spectra"
- **in prep:** Janin E. et al., Modeling Atmospheric Lines By the Exoplanet Community (MALBEC): An Intercomparison of Radiative Transfer Tools for Transmission Spectroscopy

📄 Co-author

- Fisher T., Janin E., Walker S. (2025) "A Complex-Systems Approach to Exoplanet Atmospheric Chemistry: New Prospects for Ruling Out the Possibility of Life-As-We-Know-It". *Planetary Science Journal*, 6 (5), 116.
- Slocombe L., Siebert G., Janin E. et al. (2025) "Practical Assembly Theory, Including Protocols, and a Python Toolkit". In review in *Nature Protocols*
- Brooks C., Janin E. et al. (2025) "A Firefly-inspired Model for Deciphering the Alien". In review in *PNAS Nexus*. arXiv:2511.06139
- Fisher et al. (2024) "Network and Kinetics-based Biosignatures: Implications for the Putative Habitable World Observatory Design". arXiv:2501.04737
- Villanueva et al. (2024) "Modeling Atmospheric Lines By the Exoplanet Community (MALBEC) protocol version 1.0". *The Planetary Science Journal*, 5(64)
- Barnes et al. (2025) "History and Habitability of the LP 890-9 Planetary System". *The Planetary Science Journal*, 6(1), 25
- Edwards et al. (2023) "Characterising a World Within the Hot Neptune Desert: Transit Observations of LTT 9779 b with HST WFC3". *The Astrophysical Journal*, 166(4), 158
- Fauchez et al. (2021) "TRAPPIST Habitable Atmosphere Intercomparison (THAI) Workshop Report". *The Planetary Science Journal*, 2(3), 106-134
- Papineau et al. (2025) Authigenic biosignatures of microbial biomass preserved as carbonate ooids formed by abiotic chemically oscillating reactions throughout Earth history". *Geological Society of America Bulletin*, 138 (3-4), 1795-1815

ACADEMIC PRESENTATIONS

👤 Talks

Assembly of Planetary Atmospheres: An Earth-Agnostic Method for Life Detection on Exoplanets from Infrared Spectra (May 2026 – AbSciCon) + session organizer and chair

Exploring Structural Phase Transitions in Language with Assembly Theory (Apr. 2026 – EvoLang XVI)

Sagan Exoplanet Workshop

📅 July 2023

📍 Remote

AbGradCon

📅 May 2023

📍 La Jolla, CA, USA

Open-Access Exoplanet Modeling & Analysis Tools

📅 Feb. 2023

📍 Remote

VPlanet LP 890-9 simulations

📅 Sept. 2022

📍 Remote

ARIEL Science Consortium

📅 June 2022

📍 Paris, FR

The future role of Academia in the Space sector

📅 Nov. 2021

📍 London, UK

ARIEL Science Consortium

📅 Nov. 2021

📍 London, UK

Standards of Evidence for Biosignatures

📅 July 2021

📍 Remote

TRAPPIST Habitable Atmosphere Intercomparison (THAI)

📅 Sept. 2020

📍 Remote

COMPUTING SKILLS

TauREx (forward model and retrieval)

Planetary Spectrum Generator (PSG)

AssemblyTheoryTools

GGChem

VPlanet atmos

Python

Matlab

LEADERSHIP POSITIONS

UCL Space Society

President 2021 – 2022

Head of Philosophy 2018 – 2021

UCL Centre for Outer Space Studies

Organisational Committee 2021 – 2022

UCL Space Domain

Organisational Committee 2021

UCL SciCon

Event officer 2020 – 2021

UCL Francophone Journal

President 2019 – 2020

ConScience

Founder and President 2016 – 2019

Talks and 'science shows' in underprivileged French schools

Quantifying Atmospheric Complexity for Life Detection on Exoplanets using Assembly Theory and Infrared Spectra (Mar. 2026 – AASTCS 11)

Investigating HWO's Red Limit (Jan. 2026 – AAS 247th meeting)

Proxima Kósmos: Magikos as a New Paradigm for Life Detection (Oct. 2025 – Powerhouse Arts, NY)

Searching for Life as We Don't Know It: Detecting Signatures of Chemical Selection in Exoplanet Atmospheres (June 2025 – AAS 246th meeting)

Magikos: Unveiling a Speculative Technosphere using Assembly Theory (May 2025 – Proxima Kósmos Science Convening)

Proxima Kósmos: A Speculative Exploration of Life Beyond Earth (March 2025 – Getty PST Open House)

Searching for Life-As-We-Don't-Know-It: Detecting Signatures of Chemical Selection in Planetary Atmospheres (Nov. 2024 – APEX Seminar)

Assembly of Planetary Atmospheres: A New Conceptual Narrative for the Characterization of Habitable Planets (June 2024 – AbSci-Con)

Recognizing Alien Languages: How Astrophysics and Linguistics come together (Oct. 2023 – SLSA 2023)

Assembly of Planetary Atmospheres: A Complex Systems Approach to the Search for Extraterrestrial Life (Sept. 2023 – ELSI Seminar)

Assembly Theory: Why and How it will impact Astrobiology (Apr. 2023 – AZ Astrobio Symposium)

Water and Hydrogen: A promising planetary couple (June 2022 – UCL MAPS Talks)

Biosignatures: Past, Present, Future (Oct. 2021 – APEX Seminar)

Interdisciplinarity in Science: Insights from Astrobiology (Apr. 2021 – UK National Natural Sciences Conference)

CO₂ in Terrestrial Exoplanet Atmospheres (Oct. 2019 – APEX Seminar)

👥 Posters

Searching for Life as We Don't Know It: Detecting Signatures of Planetary-scale Selection in Exoplanet Atmospheric Spectra (July 2025 – HWO Conference)

Assembly of Planetary Atmospheres: Towards a New Agnostic Framework for Data Interpretation and Life Detection (June 2024 – Exoplanets 5)

Assembling Planetary Atmospheres: Towards an Agnostic Framework for Life Detection on Exoplanets (May 2023 – AbGradCon)

LANGUAGES

English ● ● ● ● ●

French ● ● ● ● ●

Spanish ● ● ● ● ●

OUTREACH

Podcast Interview (Event Horizon) about the search for life on exoplanets and complexity-based approaches (Mar. 2026)

Classroom Interventions about astrobiology, exoplanets, alien communication, assembly theory, etc. (>20) (2022 - Ongoing)

Beyond Center For Fundamental Concepts in Science: Student Helper, organizing cross-department themed discussions connecting science and philosophy (2023)

SESE Prison Education Program, developing geology and astrobiology curricula for incarcerated juveniles (2022 - 2023)

Creation of the website AstroMatters (Summer 2021)

Missions to Mars at Your Universe Festival (Mars 2020)

Physics Film Makers at UCL (2020)

ADDITIONAL COURSES

Complex Systems Summer School

📅 June-July 2023 📍 Santa Fe Institute

EANA Spring School

Hydrothermal Vents

📅 May 2021 📍 Remote

Princeton Summer School

Physics of Life

📅 June 2020 📍 Remote

Amity Astrobiology Course

📅 Fall 2020 📍 Remote

OTHER

Skydiving Coach & Tandem/Formation Videographer

UCL Competitive Rowing: 2021 Captain's Award

Competitive Track & Field: focus on cross-country and steeple-chase

Mountaineering

Amateur Archaeologist: passionate about Precolumbian Civilizations